



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across demographics, product categories, and subscription behavior.

Dataset at a Glance

3,900

Purchases

18

Columns

37

Missing Values

Review Rating column

\$59.76

Avg Purchase

Features span demographics, purchase details, and shopping behavior — including discount usage, promo codes, and review ratings.

Data Preparation in Python

→ Clean & Standardize

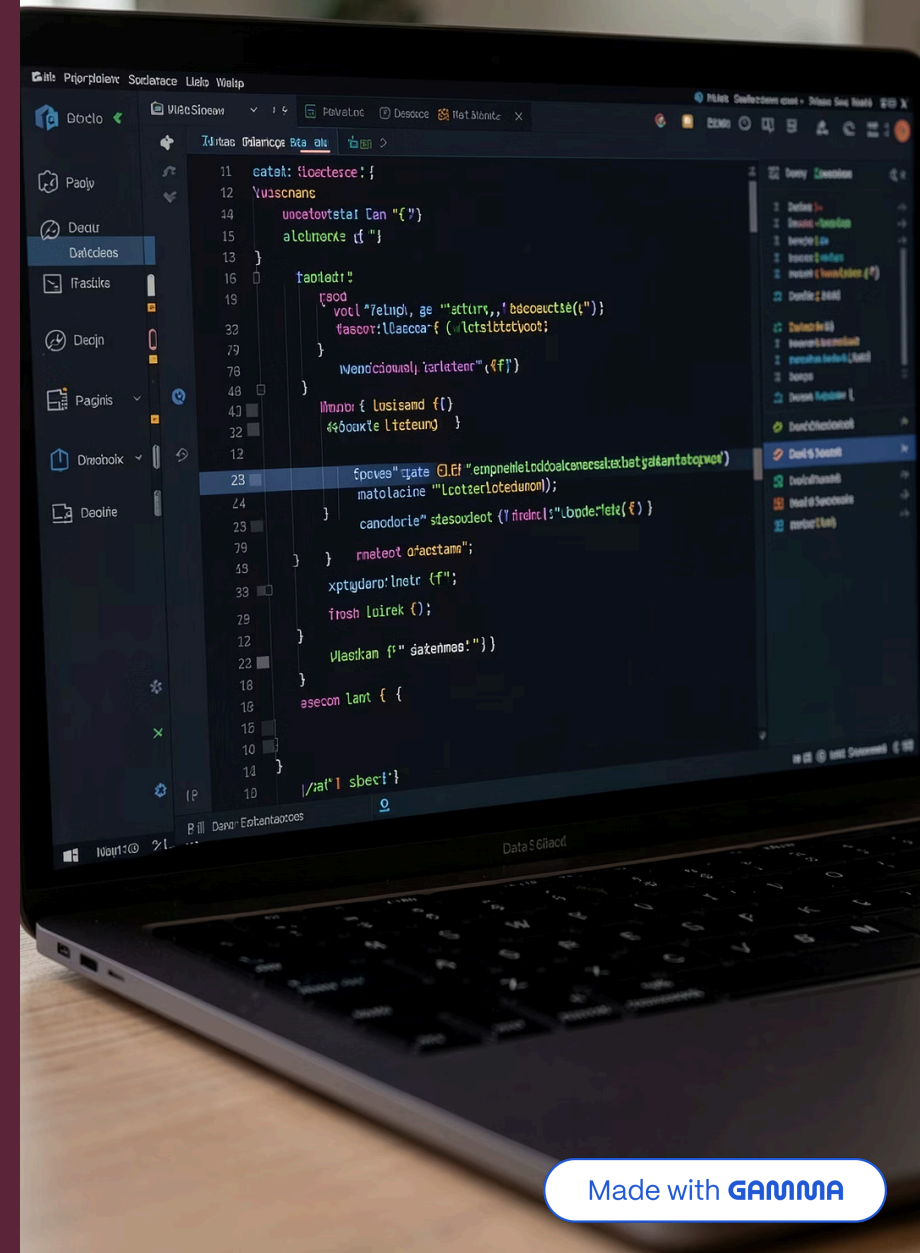
Snake_case columns, median imputation for missing ratings

→ Feature Engineering

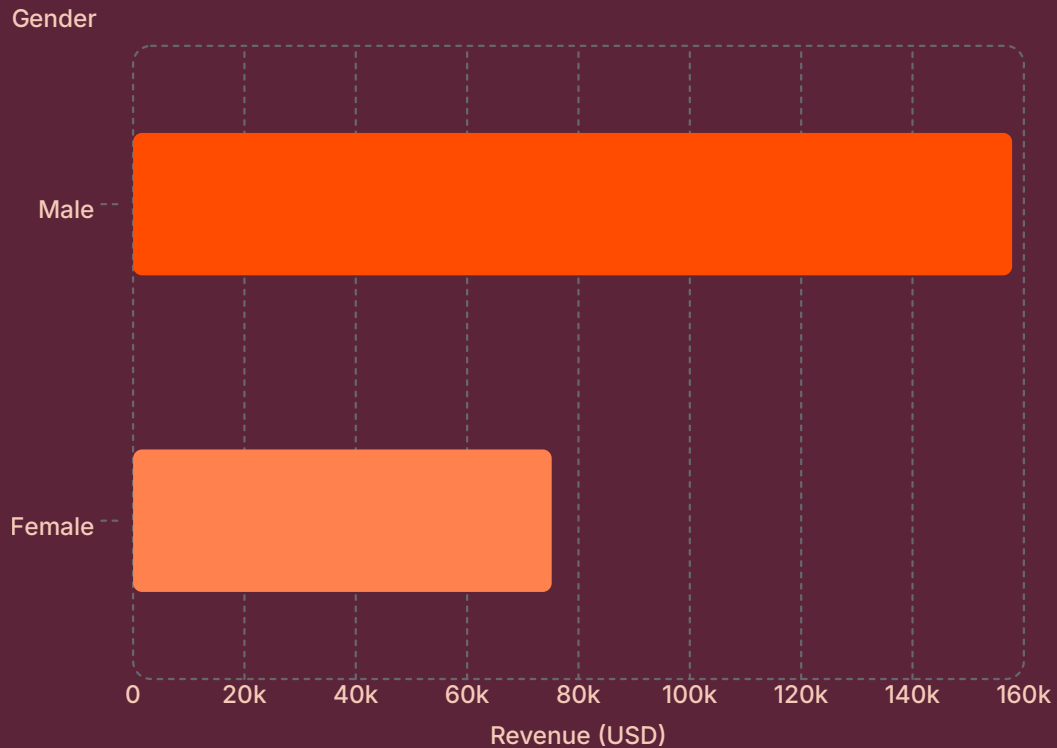
Created **age_group** and **purchase_frequency_days** columns

→ Database Load

Cleaned DataFrame loaded into MySQL for SQL analysis



Revenue by Gender



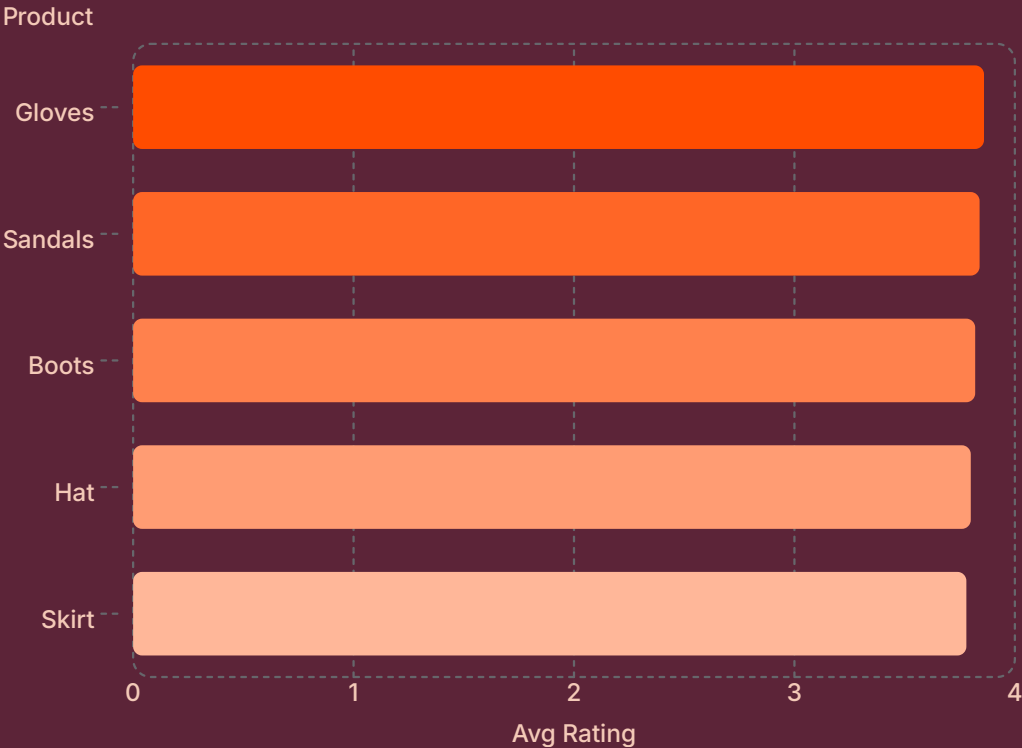
Key Insight

Male customers generated **2.1x more revenue** than female customers — \$157,890 vs \$75,191.

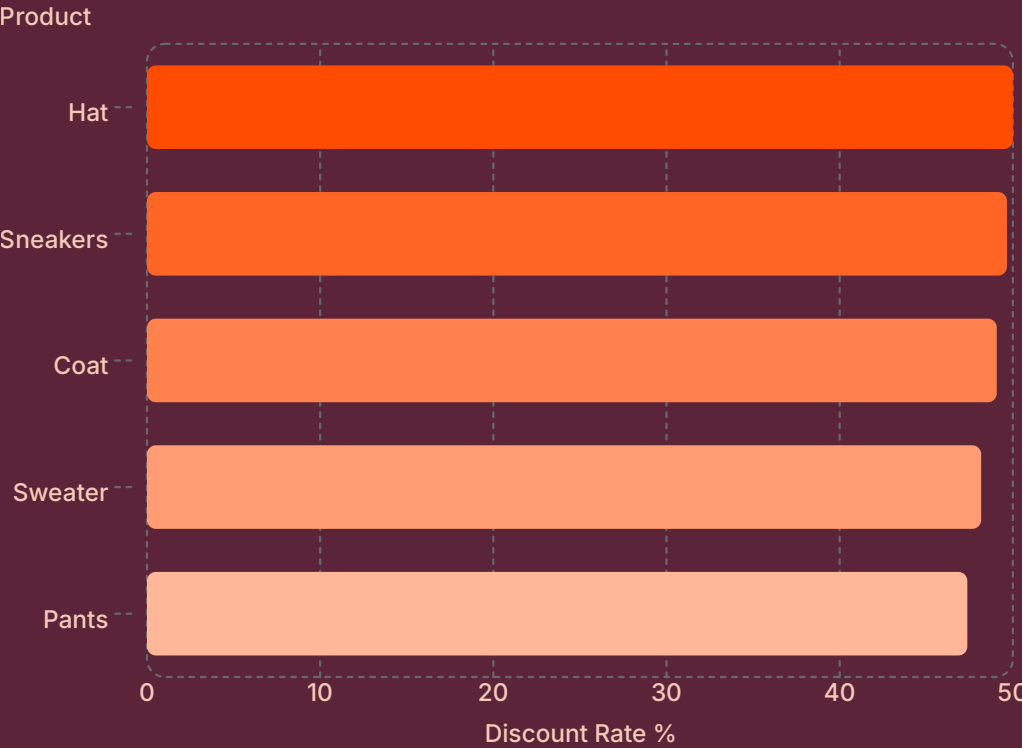
✓ Male shoppers account for 68% of total revenue.

Top Products by Rating & Discount Rate

Highest Rated Products

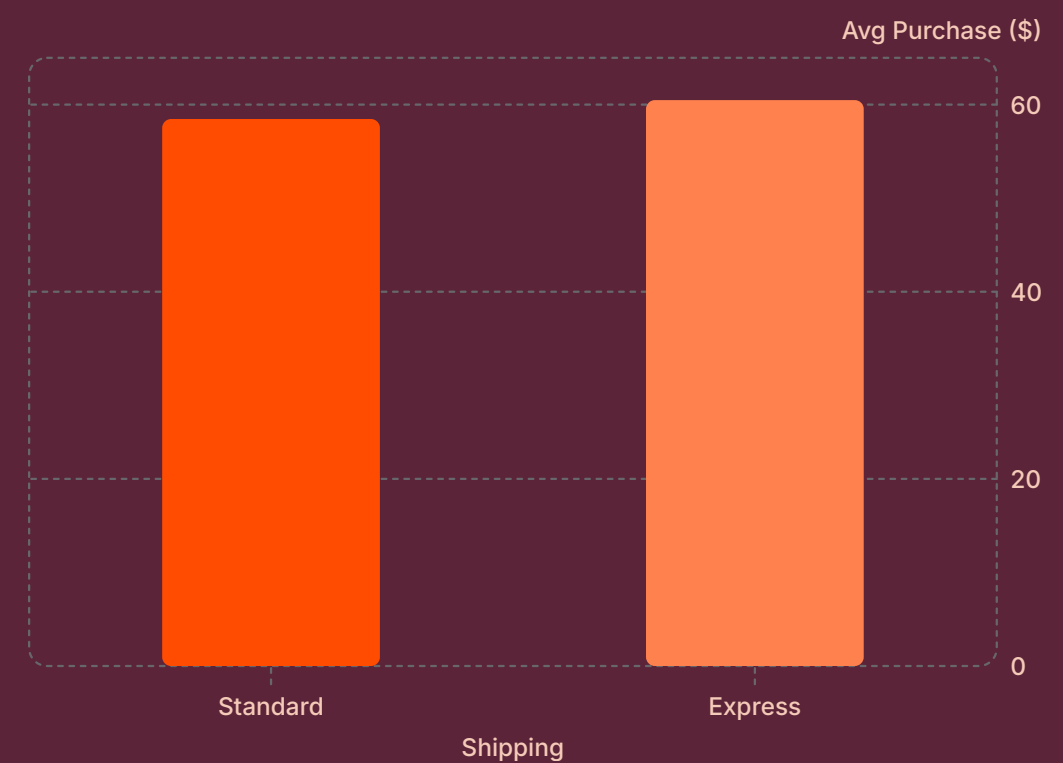


Most Discounted Products

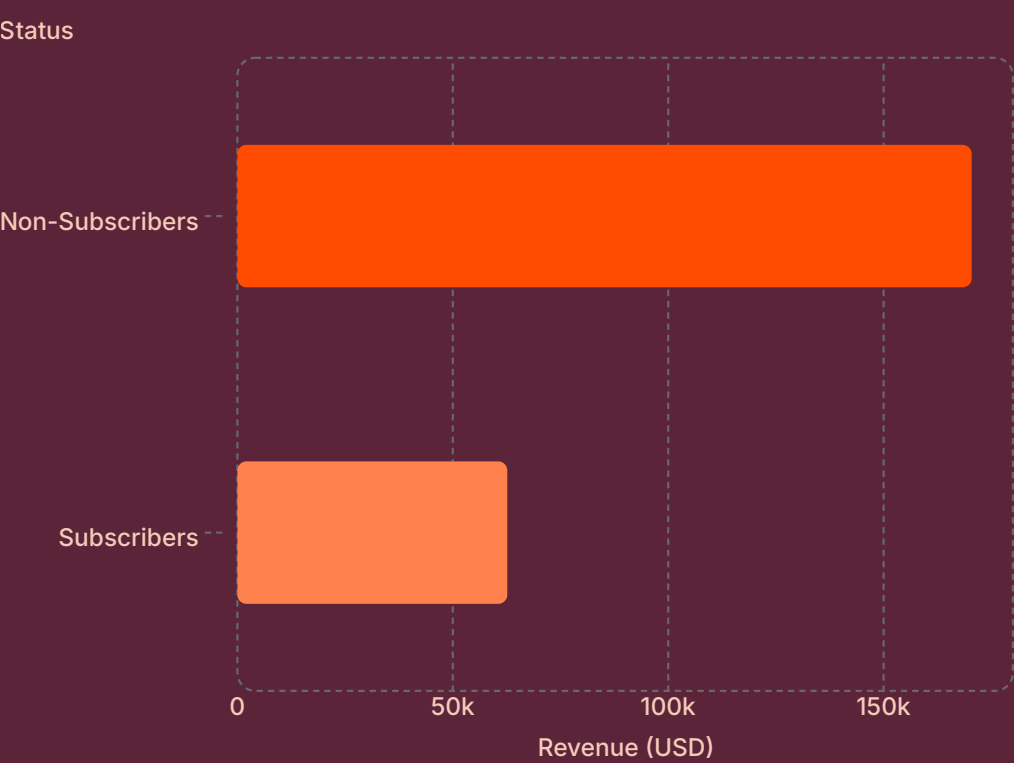


Shipping & Subscription Insights

Avg Purchase by Shipping Type

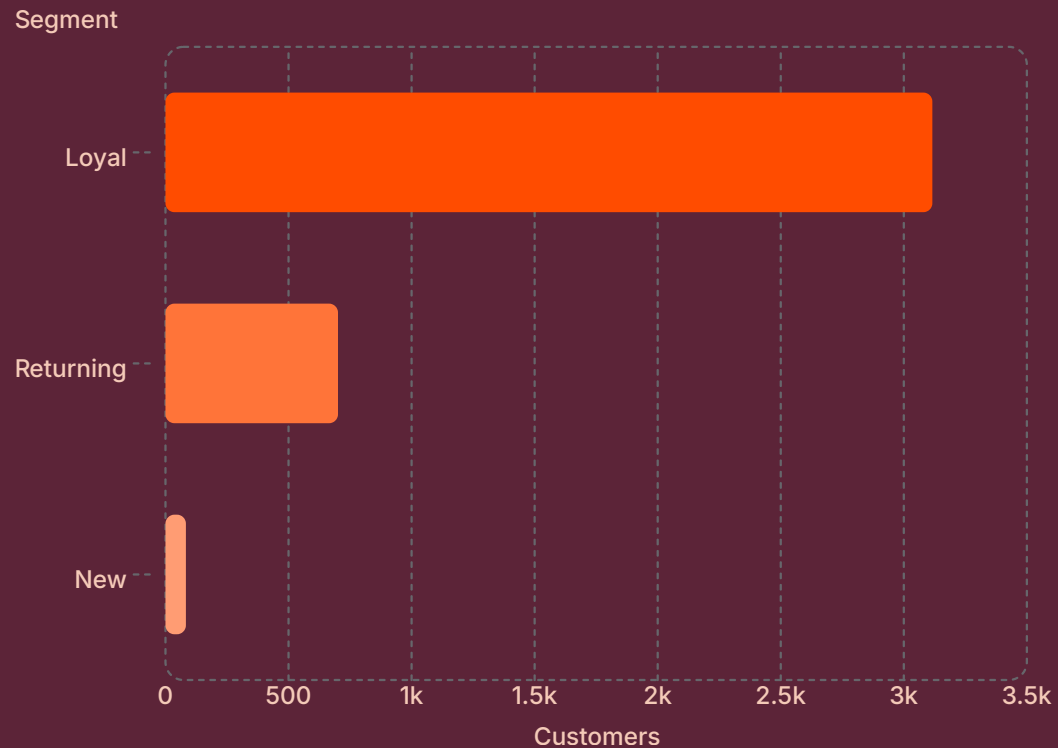


Subscribers vs Non-Subscribers



Express shipping customers spend slightly more. Non-subscribers drive **73% of total revenue**.

Customer Segmentation

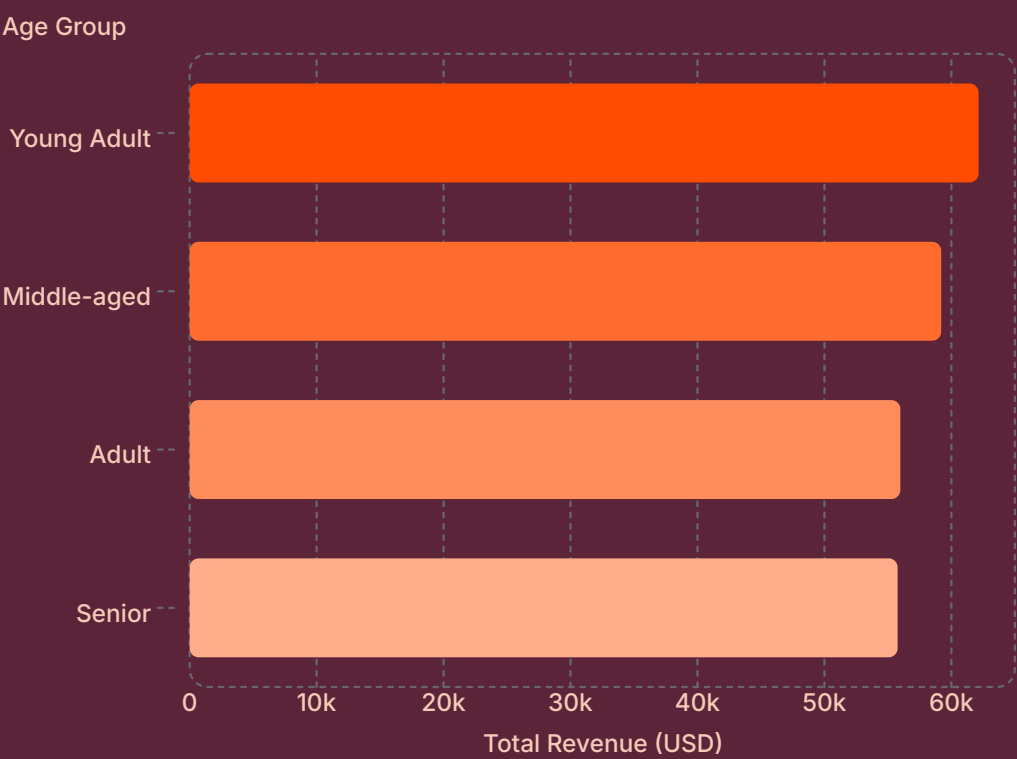


Segment Breakdown

- **Loyal** — 3,116 customers (80%)
- **Returning** — 701 customers (18%)
- **New** — 83 customers (2%)

Of customers with 5+ purchases, 958 are subscribers vs 2,518 non-subscribers.

Revenue by Age Group



Top Category: Clothing

Blouse, Pants, and Shirt lead with ~170 orders each.

Footwear Leaders

Sandals (160), Shoes (150), Sneakers (145)

Power BI Dashboard

KPIs

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Filters

Subscription, Gender, Category, Shipping Type

Visuals

Revenue & sales by category and age group

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers → Loyal segment

Review Discount Policy

Balance sales lift with margin control

Product Positioning

Highlight top-rated, best-selling items

Targeted Marketing

Focus on high-revenue age groups & express shippers